

# REPORT: NETWORK PROTOCOL FUNCTIONS

| PROTOCOL                                          | FUNCTION                                                                                                                                  | PURPOSE IN OUR LAB                                                                                                   |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>IP (Internet Protocol)</b>                     | Provides logical addressing (IP addresses) to identify devices on a network. This is the foundation of routing and communication          | Used to identify and reach PCs and router interfaces. We configured these manually and via DHCP.                     |
| <b>DHCP (Dynamic Host Configuration Protocol)</b> | Automates the assignment of IP addresses, subnet masks, default gateways, and other network configuration to devices                      | Our router acted as the DHCP server, assigning addresses to client PCs without manual input.                         |
| <b>DNS (Domain Name System)</b>                   | Translates human-readable domain names (e.g., <code>myserver.local</code> ) into numerical IP addresses that computers use to communicate | The DNS server allowed PCs to ping <code>myserver.local</code> instead of an IP address, simplifying network access. |